MARUTI SUZUKI

XCELerate 2026

Powering the Future of Mobility!









Platform Partner unstop

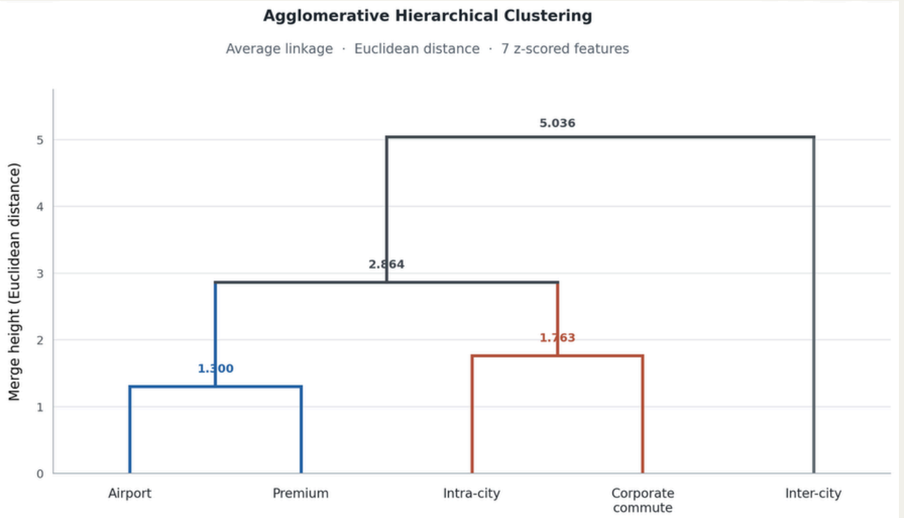
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THESIS

WHAT MARKET SAYS?

USE CASE	AVG KM	KM/ DAY	TRIPS/ DAY	₹/TRIP	GROWTH	EV SHARE	PRICE SENS.
Intra-city street hail	12.7	95	7.5	350	18%	10%	0.85
Daily commute / subscription	22.0	176	8.0	550	24%	12%	0.90
Airport transfer	35.0	210	6.0	600	26%	18%	0.35
Inter-city / outstation	280.0	340	1.5	4,200	20%	5%	0.40
Premium / shared-premium	18.0	144	8.0	520	28%	15%	0.30



WHERE WE PITCH?

*India's ride-service **volume** is in intra-city. India's ride-service **margin** is in airport and commute. This matters because they require fundamentally **different** vehicles.*

USE CASE	KM/DAY	GROWTH	EV SHARE	DRIVER ₹/MO	MACH1 FIT
intra-city	95	18%	10%	68,250	weak
premium	144	28%	15%	108,160	moderate
commute	176	24%	12%	114,400	STRONG
airport	210	26%	18%	93,600	STRONG
intercity	340	20%	5%	163,800	weak

THE PRIMARY POSITION
CLUSTER A — AIRPORT + CORPORATE-COMMUTE

- *TRIPS ARE PRE-BOOKED → THE VEHICLE'S DAY CAN BE PLANNED*
- *PICKUP IS A FIXED POINT → DEPOT CHARGING IS SCHEDULABLE OVERNIGHT*
- *THE DUTY IS PREDICTABLE → THE 45 KWH PACK AND THE 200 KM/DAY SIZING ARE SAFE*
- *EV SHARE IS ALREADY 18% → THE OPERATOR IS ALREADY ELECTRIFYING THIS LANE*

AIRPORT TRANSFERS ARE ALSO THE FASTEST-GROWING TRIP PURPOSE AT 19.14% CAGR, AND UBER BLACK'S TOP TWO USE CASES ARE AIRPORT AND CORPORATE TRAVEL.

WHERE WE DO NOT FIT

INTRA-CITY

- 61% OF THE MARKET, BUT WEAKEST FIT FOR E1.
- 95 KM/DAY; UNPREDICTABLE HAIL COMPLICATES DEPOT SCHEDULING.
- HIGHEST PRICE SENSITIVITY.

INTER-CITY

- 560 KM DRIVEN PER ROUND TRIP FOR 280 KM OF REVENUE.
- LOWEST EV SHARE: 5%.
- EMPTY RETURN TRIPS REDUCE EFFECTIVE UTILISATION AND WEAKEN ECONOMICS.

WHO ARE OUR AUDIENCE?

RANK	CRITERION	EVIDENCE
1	Cost per kilometre	every fleet TCO source converges here
2	Paid km/day (utilisation)	now the binding constraint under flat-fee platform economics
3	Uptime	charging downtime is the failure mode operators actually fear
4	Depot control	without controlled overnight charging, EV TCO does not hold
5	Residual value	BluSmart assumed 10%, realised 26% — our exact target segment
6	Financeability	CAPEX structure alone is worth ~8 points of EV advantage
7	Range	near the bottom — 315 km against a 200 km/day duty is ample

FLEET OPERATORS.

Every commercial conversation about a fleet vehicle ends at an operations manager's desk, and that person buys on cost per kilometre and utilisation.. Structurally, drivers now choose the platform; operators choose the vehicle.

BluSmart's exit redistributed supply toward high-utilisation,depot-controlled routes.

REPLACEMENT CAPACITY TARGETED AIRPORT AND CORPORATE TRAVEL:

1. EVERA EXPANDED AT DELHI TERMINALS
2. GMR HYDERABAD LAUNCHED A 200+ VEHICLE PREMIUM AIRPORT FLEET
3. BHARAT TAXI LAUNCHED AT NAVI MUMBAI.

VEHICLE CONCEPT PROPOSAL

Powertrain & Energy

FLEETLINE E1 FEATURE	WHY A FLEET OPERATOR NEEDS IT
SPEC 45 kWh LFP pack, semi-structural CTP	200 km/day duty with 10% reserve held at year 8
SPEC PMSM 80 bhp, single-speed reduction drive	No shift loss across 62,400 km/yr, and no gearset to maintain
SPEC Liquid-plate cooling, 48 °C cell design point	Holds degradation inside the validated band — passive cooling does not
ADD ⚑ Per-cell pack SoH telematics, over-the-air	DECISIVE — THE ONE THAT EARNS THE WARRANTY. Published LFP studies disagree 3.6× on time-to-80%, so a capacity-retention guarantee is only credible if the pack reports its own degradation

Safety & Liability

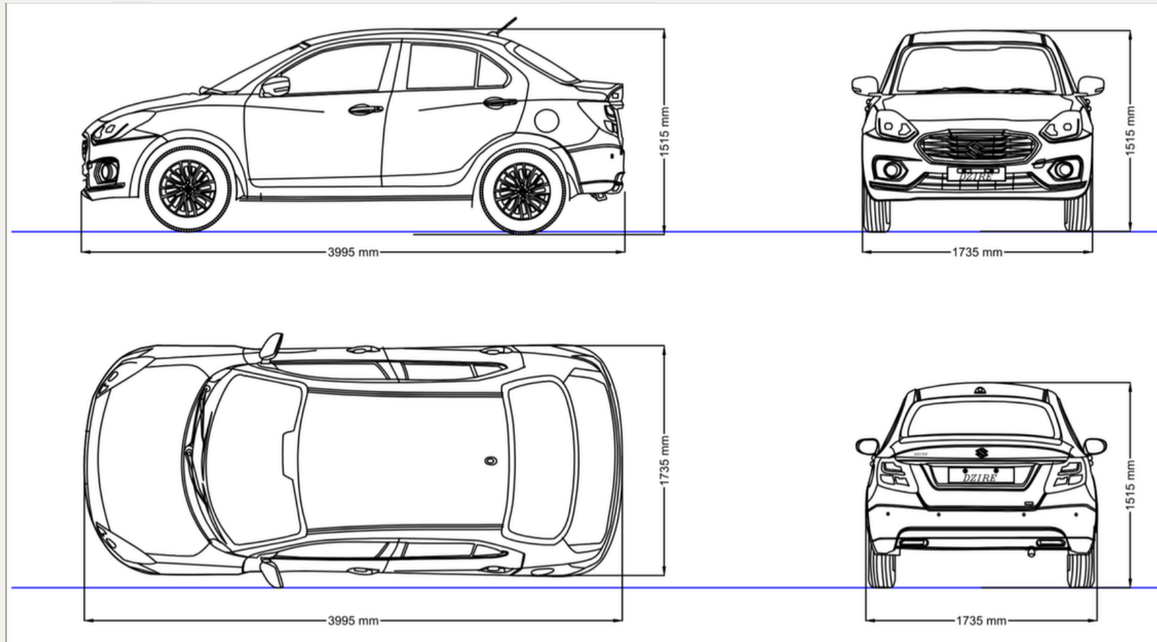
FLEETLINE E1 FEATURE	WHY A FLEET OPERATOR NEEDS IT
SPEC L2 ADAS, six airbags, ABS with EBD + CSC	Corner Stability Control matters more on a tall, pack-loaded sedan
ADD 360° HD camera with 2D/3D views	Blind-spot elimination at low speed, and dense-city parking without a spotter
ADD Dashcam recorder (DVR)	10+ paid hours a day — onboard evidence for disputes, when the driver's account is not enough

Uptime & Reliability

FLEETLINE E1 FEATURE	WHY A FLEET OPERATOR NEEDS IT
ADD Intelligent tyre pressure monitoring	Turns a roadside blowout into a scheduled wheel change. Tyre failure is a leading cause of lost paid hours

Driver Working Conditions

FleetLine E1 feature	Why a fleet operator needs it
ADD 65 W USB-C front + rear wireless phone deck	Two riders, two phones, one shift, no cable hunt between fares
ADD Ventilated front seats	A 12-hour shift is the working condition, not a luxury — heat and fatigue show up as unpaid kilometres



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A SECOND CHEMISTRY.
A SOVEREIGN SUPPLY CHAIN.
A CALCULATED BET ON 2035.

THE HERETIC'S THESIS

THE INDUSTRY HAS A DEFAULT. WE HAVE A DIFFERENT THESIS.
SODIUM-ION NEED NOT DEFEAT LITHIUM-ION TODAY. IT NEEDS TO BECOME A CREDIBLE ALTERNATIVE BY 2035. MACH1 IS DESIGNED TO GIVE IT THAT CHANCE—WITH LFP AS THE FALLBACK AND THE FINAL CHEMISTRY DECIDED BY COST, SUPPLY SECURITY AND DEMONSTRATED PERFORMANCE.

- India's battery problem is not just cost – it is **control.** Today, India's entire EV push runs on **lithium-ion** cells manufactured and processed almost entirely **outside the country**. The Ministry of Heavy Industries has formally acknowledged this **dependency**, identifying supply-chain concentration and lithium price volatility as strategic risks. Sodium-ion is not proposed as a superior chemistry. It is proposed as a second chemistry – one that reduces India's exposure to a single mineral base, enables aluminium current collectors in place of copper, and opens a **domestically controllable** supply chain. The IEA's 2026 Global EV Outlook confirms this framing directly: **sodium-ion's** lithium **independence** is already reinforcing industrial interest globally, even as the technology scales.
- The 2035 horizon is what makes this viable rather than speculative. Current sodium-ion manufacturing capacity sits at just **over 1% of lithium**-ion – the technology is not ready today. But CATL and BYD are both **developing** it at scale, announced projects for 2030 represent roughly 7% of committed lithium-ion capacity, and sodium-ion cells now retain **~90% capacity** at -40°C with validated operation up to 70°C – directly relevant to Indian conditions. MACH1 **does not bet** on sodium-ion winning. It bets on having enough runway to watch it mature, with **LFP** as the confirmed **fallback**. If sodium-ion crosses the cost and supply gate by **2032–33,** it drops in. If it doesn't, **LFP launches**. The architecture supports both. The **risk is managed.**

PUBLIC FUNDING IS MOVING SODIUM-ION RESEARCH TOWARDS COMMERCIALISATION

May 2025 — JNCASR develops a fast-charging sodium-ion cell

The Department of Science and Technology (DST) publicised research by the Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR), reporting a sodium-ion battery that charged to 80% in six minutes and achieved more than 3,000 charge cycles in the reported testing.

December 2025 — ARCI moves into industrial cell validation

The International Advanced Research Centre for Powder Metallurgy and New Materials (ARCI), an autonomous R&D institution under DST, signed an agreement with Voltasun Technologies on 15 December 2025 to supply 80 sodium-ion pouch cells rated at 5 Ah each for industrial-performance evaluation.

December 2025 — Government-backed Indo-Israel industrial R&D

The Technology Development Board (TDB), under DST, entered into an agreement with NOPO Nanotechnologies India under the India–Israel Industrial R&D and Technological Innovation Fund. The project investigates integrating single-walled carbon nanotubes into sodium-ion batteries and developing a proof-of-concept prototype.

April 2026 — Direct support for commercial-scale sodium-ion materials

TDB announced financial assistance to Indigenous Energy Storage Technologies Pvt. Ltd. for commercialisation of hard carbon derived from agricultural and other biomass waste, intended for use as a sodium-ion battery anode material.
This is strategically important because hard carbon is a key sodium-ion input, and the IEA identifies its supply chain as underdeveloped and heavily concentrated in China

9th September 2026 — NTPC issues a sodium-ion grid-storage pilot EOI

This is one of the clearest current policy signals. NTPC's official tender portal records an Expression of Interest titled "EOI for Setting up A Pilot Project for Development and Installation of a Grid-Connected BESS using Sodium-Ion Battery Technology."

BROADER NATIONAL POLICY IS CREATING THE CONDITIONS FOR BATTERY-CHEMISTRY DIVERSIFICATION

TOGETHER, THESE INITIATIVES DEMONSTRATE A NATIONAL EFFORT TO REDUCE DEPENDENCE ON IMPORTED BATTERY INPUTS AND BUILD DOMESTIC MANUFACTURING CAPABILITY. SODIUM-ION IS RELEVANT TO THAT OBJECTIVE BECAUSE IT OFFERS AN ALTERNATIVE PATHWAY WITH A DIFFERENT RAW-MATERIAL PROFILE.

NATIONAL PROGRAMME ON ADVANCED CHEMISTRY CELL BATTERY STORAGE.

The Ministry of Heavy Industries' PLI scheme has an approved outlay of ₹18,100 crore to build gigascale advanced-chemistry-cell manufacturing in India, with a stated emphasis on domestic value addition. The scheme provides an industrial framework for building battery manufacturing capabilities

NATIONAL CRITICAL MINERAL MISSION

Approved in January 2025, the mission provides for ₹16,300 crore in public expenditure over seven years, with an expected ₹18,000 crore of investment from public-sector undertakings and other stakeholders. Its purpose is to strengthen domestic critical-mineral value chains across exploration, mining, processing and recovery.

ICEA ENERGY STORAGE REQUIREMENTS

India's Central Electricity Authority has estimated an energy-storage requirement of approximately 411 GWh by 2031–32, driven by the need to integrate renewable electricity reliably. The figure was reported in a December 2025 parliamentary response published by the Ministry of Power.

INDIA IS PURSUING BATTERY SELF-RELIANCE AND CRITICAL-MINERAL DIVERSIFICATION; SODIUM-ION IS EMERGING AS ONE TECHNOLOGY CAPABLE OF CONTRIBUTING TO THOSE GOALS.

WHY 2035 DOES NOT SOLVE IT

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This is not a naive claim, and it deserves a fair hearing. Sodium-ion's cost trajectory is real, and layered-oxide sodium cells genuinely do reach LFP price parity on a five-year view. The instinct — "we launch in 2035, so wait for the better chemistry" — is commercially reasonable. ***IT FAILS ON TIMING AND CERTIFICATION, NOT ON DIRECTION.***

WALL 1 - AVAILABILITY AT SCALE

THE CELL DOES NOT EXIST AT OUR SCALE. NO SUPPLIER SHIPS LAYERED-OXIDE SODIUM-ION IN A SEDAN-CLASS VEHICLE. CURRENT AUTOMOTIVE DEPLOYMENTS ARE A00-SEGMENT MICROCARS AT 23–25 KWH (HINA IN THE JAC YIWEI 3, FARASIS IN THE JMEV EV3). THE ONE SODIUM-ION PRODUCT WITH GWH CAPACITY AND A PASSENGER CAR AND GB 38031-2025 CERTIFICATION IS CATL NAXTRA — AND IT IS PRUSSIAN WHITE, A DIFFERENT CATHODE FAMILY FROM THE LAYERED OXIDE THE PROPOSAL SPECIFIES. PRUSSIAN WHITE IS NOT A DROP-IN SUBSTITUTE FOR THE SPECIFIED CHEMISTRY.

WALL 2 — IT CANNOT BE LEGALLY HOMOLOGATED IN INDIA.

AIS-038 REV.2 AMENDMENT 3, ANNEX IX-K(2), VERBATIM:
"CELLS USED TO MAKE REESS, SHALL BE CERTIFIED AS PER IS 16893-PART 2 AND PART 3 BY NABL ACCREDITED LAB OR BY TEST AGENCY NOTIFIED UNDER CMV RULE 126."
IS 16893-2 AND -3 ARE TITLED "SECONDARY LITHIUM-ION CELLS…". NO AMENDMENT THROUGH AMD 4 HAS ADDED AN ALTERNATIVE-CHEMISTRY CLAUSE OR SCOPE EXEMPTION. NO SODIUM-ION CELL HAS EVER BEEN CERTIFIED TO THEM.

DECISION

LFP IS THE PRIMARY CHEMISTRY. SODIUM-ION IS A DOCUMENTED 2036–38 SECOND ACT. RETAINED PRECISELY BECAUSE THE ARCHITECTURE KEEPS IT AVAILABLE. WE ARE NOT ARGUING SODIUM-ION IS THE WRONG DIRECTION. WE ARE ARGUING IT CANNOT BE CERTIFIED ON THE PROPOSED DATE, AND A PROGRAMME THAT CANNOT BE CERTIFIED CANNOT BE DELIVERED.

ARCHITECTURE

WHY DETACHABLE, NOT STRUCTURAL

- *CHEMISTRY SWAP BECOMES A SOFTWARE CHANGE, NOT A NEW CAR*
- *AIS-038 CLAUSE 2.48: A CHEMISTRY CHANGE NEEDS FULL REESS RETEST EITHER WAY*
- *BUT DETACHABLE MAKES THE CELL SWAPPABLE. STRUCTURAL CTC LOCKS THE CHEMISTRY FOREVER*
- *COST OF THAT OPTIONALITY: ₹2–3 CR / 4–6 MONTHS VS ₹15–25 CR / 18–24 MONTHS*

Dimension	Detached semi-structural CTP	Full structural CTC
Chemistry change	Cell swap + firmware	Effectively impossible
AIS-038 Clause 2.48 retest	4–6 months, 2–3 Cr	18–24 months, 15–25 Cr
Cell standardisation	256×106×32 prismatic, 102–108 s	Chemistry-locked
Repurposing / second life	Straightforward	Very difficult
Body programme impact	None	Full body-in-white retool

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LIFE CYCLE MODEL

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$$SoH(t, EFC) = 1.02 - A_{cal} \cdot t^{\alpha} \cdot Arrhenius(T) \cdot k_{SOC} - k_{cyc} \cdot DoD^{\beta} \cdot Arrhenius(T) \cdot (EFC \cdot 8)$$

WHERE EVERY TERM IN THE EQUATION COMES FROM

- α — Capacity-fade law (dominant uncertainty): Lewerenz (2017) measured linear fade across 50 LFP cells ($\alpha=1.00$); Naumann (2018) fitted √t fade ($\alpha=0.50$), consistent with Gasper (2021, ~0.45) and Liu (2018, 0.42). This 2.3× spread shifts the estimated pack requirement from ~44 to 76 kWh.
- β — DoD dependence of resistance growth: Gauthier (2022) reports $\beta=+1.0$ at 40°C; Eldesoky (2022) finds no DoD dependence below a 4.06 V cut-off; Eldesoky (2020) reports $\beta=-0.40$ at 55°C. These competing findings shift pack sizing by ~7 kWh.
- Engineering implication: The uncertainty bands trace to two specific literature disagreements—not arbitrary model assumptions. α is the dominant uncertainty.
- Validation strategy: The proposed cell-testing programme targets both parameters rather than splitting the difference between conflicting studies.

Symbol	Term	Value	Source	Status
1.02	cal · offset	1.02	fitted intercept	locked
- A_cal	cal · coefficient	—	Naumann 2018	locked
· t^α	cal · time exponent	**0.42 → 1.00**	Lewerenz '17 · Naumann '18 · Gasper '21 · Liu '18	**▲ CONTESTED — 32 kWh pack**
· Arrhenius(T)	cal · temperature	Ea 17.126 kJ/mol (capacity)	Naumann 2018	locked
· k_SOC	cal · SOC factor	0.6140 @ 66% mean SoC	Naumann 2018 (literal form)	locked
- k_cyc	cyc · coefficient	—	Naumann 2018	locked
· DoD^β	cyc · DoD exponent	**-0.40 → +1.00**	Gauthier '22 · Eldesoky '22 · Eldesoky '20	**▲ CONTESTED — 7 kWh pack**
· Arrhenius(T)	cyc · temperature	Ea 71.827 kJ/mol — **4.2× the capacity Ea**	Naumann 2018	locked
· EFC · 8	duty · full-charge count	~200 EFC/yr → 1548 @ yr 8	duty: 12.84 kWh/100 km · 25.68 kWh/day	locked

WHAT THE EQUATION PREDICTS

- Uncertainty budget: Table 1's contested parameters account for the full band width: α contributes 32 kWh; β contributes 7 kWh. The spread reflects unresolved literature disagreements, not arbitrary model uncertainty. We use finite, specific defaults at each temperature rather than averaging conflicting studies.
- The thermal limit: At 55°C, the worst-case estimate exceeds the model's 140 kWh search ceiling—roughly 3× the 45 kWh sedan baseline. We retain the ceiling rather than force a misleading finite answer.
- Design decision: Specify a maximum 48°C cell temperature under liquid-plate cooling. This avoids the unresolved worst-case corner and keeps pack sizing within a practically buildable range.

Cell temp	Pack required (band)	Pack @ defaults	SoH @ yr 8 (band)	SoH @ defaults	R_inc @ yr 8	Verdict
26 °C	37.1–53.3 kWh	37.2 kWh	59.5–85.5%	85.2%	+14.0%	✓ Wide margin
40 °C	40.6–74.6 kWh	41.0 kWh	42.5–78.1%	77.3%	+50.9%	✓ Achievable
48 °C ★	43.3–100.1 kWh	44.2 kWh	31.7–73.2%	71.8%	+101.3%	▲ Band too wide
55 °C	46.2–**140+** kWh	47.8 kWh	22.0–68.6%	66.4%	+179.8%	× Infeasible

DISAGREEMENTS

We built this from published cell data, not vendor claims. We found real disagreements in the literature and we showed them to you. And we can tell you exactly which test closes each gap."

01 — PACK DESIGN

Proposed architecture

- Chemistry: Prismatic LFP cells; lower material-cost exposure and established cycle-life evidence.
- Configuration: Detachable, semi-structural cell-to-pack (CTP); standardised pack envelope, common BMS hardware and chemistry-specific firmware.
- Thermal management: Active chiller-loop cooling as the design reference. Liquid-plate cooling is modelled at a higher 48°C cell temperature;
- Sizing: Model baseline is 41.3 kWh at 40°C and 44.2 kWh at 48°C. These are conditional estimates—not a final pack specification.

02 — DUTY CYCLE

	EV (PROPOSED)
ENERGY CONSUMPTION	12.84 kWh/100 km
DAILY DISTANCE	200 km
DAILY ENERGY DEMAND	25.68 kWh
ANNUAL EQUIVALENT FULL CYCLES (EFC)	190–228; ~200 nominal

03 — TOTAL COST OF OWNERSHIP (TCO)

LOCATION	MACH1(EV)	CNG SEDAN	EV ADVANTAGE
DELHI	₹2.55/km	₹3.15/km	19%
BANGALORE	₹2.80/km	₹3.60/km	22%
MUMBAI	₹2.95/km	₹3.80/km	22%
HYDERABAD	₹2.65/km	₹3.35/km	21%

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TECHNICAL SPECIFICATIONS

POWERTRAIN	80 bhp continuous / 100 bhp peak - PMSM - Water-cooled Peak + short duration, continuous rating derates at 45°C ambient
DRIVE	Single-speed reduction - BEV
MOTOR EFFICIENCY	93–97% peak - –2% drop at 45°C
BATTERY / RANGE	LFP - 45 kWh - Structural pack 10–90% - 36 kWh usable 3% top and bottom reserved for cycle-life protection
RANGE	263 km claimed - 200 km @ 75% DoD (feet (operating range)
DAILY ENERGY	–25.7 kWh @ 200 km/day - –20–35% with AC load at 45°C
PACK THERMAL MGMT	Active liquid cooling - pack maintained ≤45°C
CHARGE RATE	1C max - 45 kW DC fast charge - CC–CV protocol
DEPOT CHARGING	11 kW AC - 3-phase - –23 hr overnight window
RATED LIFE	200,000+ km - –3.2 yr at 62,400 km/yr duty cycle
BRAKING / CHASSIS	Front: Disc - Rear : Drum ABS - ESC
WHEELS / TIRES	14' alloy - 175/56 R14 tubeless radial
GROUND CLEARANCE	170 mm unladen
SAFETY / COMPLIANCE	**Airbags - ABS - Level 2 ADAS** **AIS-156 - AIS-038 Rev.2 - ECE R100**
BMS	Cell-level voltage + temperature monitoring
INTERIOR / COMFORT	5-seat - 2row - rear-biased - flat floor (no tunnel) Rear AC vents - durable leatherette - ventilated front seats EPS - Full LED - DRL bar - Fog lamps
TOP VARIANT	Panoramic sunroof
LUGGAGE	400 L
BATTERY FIRST-LIFE	– 15 yr (cycle-life basis)
CORROSION / NVH	Validated for coastal and monsoon deployment
PRIMARY WEAR ITEM	Bearings
SEGMENT	Compact Sedan - B-segment

3,995	1,684	1,532	2,450
LENGTH MM	WIDTH MM	HEIGHT MM	WHEELBASE MM

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POWERTRAIN STRATEGY



OPERATOR COST, 5 YEARS, 312,000 KM

	MACH1(EV)	CNG SEDAN
COST/KM	₹6.42	₹7.63
RESIDUAL	₹4.82	₹6.67
ADVANTAGE	16% / 28%	

OPERATOR COST, 5 YEARS, 312,000 KM

CNG MILEAGE	CNG COST/KM	EV ADVANTAGE
20 KM/KG	₹8.34	23%
25 KM/KG	₹7.63	16%
30 KM/KG	₹7.42	13%

"Higher kilometers traveled equates to better relative TCO of the EV option." — Cadmus / Uber, Electrifying Ridehail in India, Jan 2026

BEV. LFP. SINGLE-SPEED DRIVE.

High-mileage taxis favour EVs on total cost of ownership, with CNG as the alternative where infrastructure exists. — CEEW

200 km/day makes the economics work.

Fleet parity with diesel typically emerges at 120–200 km/day, with 18–24-month payback. Our depot-return duty cycle sits at the favourable end of this range.

Depot charging eliminates downtime.

A 36 kWh recharge (10–90%) takes 4.9 hours at 7.4 kW or 3.3 hours at 11 kW—completed overnight, with zero revenue hours lost.

Cheap, scheduled charging protects margins.

Depot electricity at ₹5.50–9/kWh beats public fast charging at ₹18–24/kWh. Smart off-peak scheduling can cut charging costs by 20–35%, with a feasible 1:3 charger-to-vehicle ratio.

01 — COMPARISON AND JUSTIFICATION

Corroborated by CEEW:
"For taxis, high annual mileage makes EVs the most cost-effective option, followed by CNG where infrastructure exists."

	BEV	HEV	PHEV	CNG	BIO CNG
COST/KM (OPERATOR)	₹6.42	₹9–11	₹8–10	₹7.42–8.34	≈ CNG + supply risk
REFUELLING READINESS	Depot AC overnight; zero revenue-hour downtime	Full — ICE base	Full — ICE base	Best-in-class	Best-in-class
RANGE / UPTIME	315 km real (verified)	Unaffected	Unaffected	~500–600 km real	~500–600 km real
SUSTAINABILITY	Best available on a coal-heavy grid	Marginal	Marginal	Methane slip; worse CO2	Genuinely low-carbon but supply immature
CHOSEN BY REGULATORS	Yes — Delhi gazette 100% by 2030	No	No	No	No

"Total cost of ownership parity with diesel often emerges when vehicles run 120–200 km daily, with payback in roughly 18–24 months... economic parity can begin at 100–150 km per day depending on tariffs and financing structures." — The Hindu BusinessLine, 26 Feb 2026, quoting EKA Mobility and Euler Motors

02 — CHARGING & REFUELING READINESS

India has roughly 35,000 public charging ports for 5.7 million registered EVs — 160 EVs per port (NITI Aayog index, via Cadmus/Uber). Public charging is therefore not an operating plan for a fleet. It is a fallback.

WHY BEV WOULD WORK?

	TIME FOR 36 KWH (10→90%)	FITS OVERNIGHT WINDOW?
7.4 KW AC	4.9 h	Yes
11 KW AC	3.3 h	Yes
22 KW AC	1.6 h	Yes

CHARGING ROUTE	COST
PUBLIC DC FAST CHARGE	₹18–24 /kWh
DEPOT, OFF-PEAK COMMERCIAL TARIFF	₹5.50–9 /kWh
DEPOT + ROOFTOP SOLAR	₹3–5 /kWh

At any of these, the vehicle is fully charged before the shift and loses zero revenue-hours to charging.

PACK SIZING HEADROOM

	MACH1(EV)
200 KM/DAY	34 kWh → 45 kWh is comfortable
263 KM/DAY	45 kWh
300 KM/DAY	51 kWh → 45 kWh would not close the day

WHAT ACTUALLY HAPPENS WITHOUT DEPOT CHARGING

"Around two-and-a-half months after... Green SM launched its electric taxi service... drivers said they are spending hours waiting for chargers or going offline to conserve battery... I sometimes come to charge my vehicle on my week-offs." — The Indian Express, 1 Sep 2026, Noida

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TOTAL COST OF OWNERSHIP ANALYSIS

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WHY NOT THE ALTERNATIVES?

BEV OFFERS THE LOWEST OPERATING COST FOR OUR 200 KM/DAY DEPOT-BASED DUTY CYCLE, WITH CHEAPER ENERGY, LOWER MAINTENANCE, AND OVERNIGHT CHARGING THAT AVOIDS REVENUE LOSS. LFP KEEPS THE BATTERY COST-EFFECTIVE AND DURABLE.

CNG — THE REAL COMPETITOR, AND IT LOSES ON COST

CNG wins refuelling readiness and has a far lower purchase price. It loses because at 200 km/day the running-cost saving is not enough to cover the ₹5.49 lakh capital premium within five years ,after which EV wins for the remaining life.

BIO-CNG — THE OPTION THAT IS NOT READY

Bio-CNG has the best carbon story of any fuel available, and it is not deliverable at fleet scale in 2035:

- 207 CBG plants commissioned as of April 2026, against a government target of 5,000 by 2030
- Only 17 of those 207 are pipeline-connected — the rest move by high-pressure cascade
- Commissioned capacity is ~2 MMTPA against a 15 MMTPA target
- Operating plants run at 20–60% of rated capacity on feedstock inconsistency
- Mandatory blending is only 1% in FY2025-26, rising to 5% by FY2028-29

HEV AND PHEV — STRUCTURALLY UNCOMPETITIVE HERE

CEEW is unambiguous:
"Petrol-electric hybrid vehicles... are significantly more expensive, mainly due to the high GST and compensation cess (at 43 per cent), making them uncompetitive in this segment against EVs."
The tax treatment alone is disqualifying. Beyond that, both inherit the ICE's full mechanical service burden — the exact cost line where EV gains its advantage — while adding a battery and inverter to a fleet that will keep both.

1. METHOD

PER THE BRIEF — PURCHASE, ENERGY, MAINTENANCE, UPTIME, RESIDUAL VALUE — FROM BOTH THE OPERATOR'S AND THE DRIVER'S VIEW.

- Costed once, never twice. Capital cost appears either as the cash on-road price or as the full EMI stream. An earlier draft of this analysis added both, which double-counted the loan principal and made the EV look 95% worse. That error is recorded in §10 because it is exactly the kind of mistake that survives into a jury room if nobody checks.
- Included: ex-showroom, GST, road tax, financing, energy (net of charging losses), maintenance (net of the EV tyre premium), insurance, driver downtime, residual credit.

EXCLUDED, WITH REASON:

	WHY?
AGGREGATOR COMMISSION (20–25%)	Same on both vehicles, so it cancels in the comparison. Excluding it also keeps this reconcilable with the deck's ₹57,200/ month gross revenue.
TOLL / FASTAG	Vehicle-class dependent, not powertrain dependent
DRIVER SALARY AS A COST	We treat it as revenue-side (fare × km), and model only lost earning time as a cost.
BATTERY REPLACEMENT	Outside the 8-year horizon at our duty

2. CONFIGURATION USED

VEHICLE	MACH1 — BEV, LFP 45 kWh, PMSM 80 bhp continuous, single-speed
COMPARATOR	Maruti Dzire Tour S (CNG)
DUTY	200 km/day × 26 days × 312 days = 5,200 km/mo · 62,400 km/yr
CONSUMPTION	12.84 kWh/100 km (verified duty cycle, battery phase
CHARGING	Depot AC overnight, 90% wall-to-battery efficiency
REVENUE	₹11/km net fare → ₹57,200/month
FINANCING	10% annual, 60 months, no down payment (deck's assumption)
HORIZON	5 years (loan) and 8 years (battery-phase horizon)
CITIES	Delhi, Mumbai, Bengaluru — different tariffs, road tax and CNG prices

3. THE COST STACK — DELHI, 5 YEARS, FINANCED

COMPONENT	EV	CNG	EV ADVANTAGE
On-road (ex-showroom + GST + road tax)	₹12,72,600	₹8,51,488	−₹4,21,112
Finance (EMI ₹29,743 / ₹18,092 × 60)	₹16,22,339	₹10,85,496	−₹6,99,077
Energy	₹2,00,304	₹9,48,480	+₹7,48,176
Maintenance	₹93,600	₹2,10,000	+₹1,16,400
Insurance	₹2,10,000	₹1,97,400	−₹12,600
Residual Credit	−₹3,30,876	−₹3,40,595	+₹9,719
Total	₹17,95,367	₹21,00,781	+₹3,05,414
Cost/Km	₹5.75	₹6.73	+14.5%

6.SENSITIVITY- WHAT VARIABLES INFLUENCE

INPUT CHANGE	EV ADVANTAGE
BASE (DELHI: ELEC ₹4.50, CNG ₹76/KG, 25 KM/KG)	**+14.5%**
CNG FUEL ECONOMY 25 → 30 KM/KG	**+7.6%**
CNG FUEL ECONOMY 25 → 35 KM/KG	**+1.9%**
CNG PRICE ₹76 → ₹90/KG	**+21.1%**
ELECTRICITY ₹4.50 → ₹8.00 (GENERIC COMMERCIAL)	**+7.1%**
ELECTRICITY ₹4.50 → ₹10.00	**+2.9%**
ELECTRICITY ₹4.50 → ₹12.00 (PUBLIC DC)	**−1.4% ✗**
ELECTRICITY ₹4.50 → ₹24.00 (PUBLIC DC RETAIL)	**−26.8% ✗**
EV RESIDUAL 26% → 36% (DECK'S OLD ASSUMPTION)	+20.6%
CONSUMPTION 12.84 → 15.38 KWH/100 KM (DECK'S FIGURE)	+12.7%
8-YEAR HORIZON	−6.4%
WORST CASE STACKED (15.38 + ₹12 + RESID 26%)	**−12.5%**
BEST CASE STACKED	**+20.6%**

4. OPERATOR TCO — THREE CITIES

CITY	BASIS	EV ₹/km	CNG ₹/km	EV ADV.
Delhi	Financed 60 mo	5.75	6.73	+14.5%
Delhi	Cash outright	4.63	5.98	+22.6%
Mumbai	Financed 60 mo	6.47	7.00	+7.6%
Mumbai	Cash outright	5.35	6.32	+15.3%
Bengaluru	Financed 60 mo	6.66	7.25	+8.2%
Bengaluru	Cash outright	5.45	6.51	+16.4%

*LEVER 1 — **FINANCING STRUCTURE NEARLY DOUBLES THE ADVANTAGE**. CASH PURCHASE GIVES +17.5% AGAINST FINANCED +8.4%, IN THE SAME CITY, ON THE SAME VEHICLE. THE DRAG IS THE INTEREST ON A ₹14 L ASSET: ROUGHLY **₹3.85 L OF INTEREST**, AGAINST ₹2.33 L ON THE CNG.*

*THAT IS A NEGOTIATION LEVER, NOT A PHYSICS ONE. A FLEET OPERATOR BUYING AT SCALE CAN NEGOTIATE A LONGER TENURE, A LOWER RATE, OR AN EV-SPECIFIC PRODUCT. **A 7-YEAR TENURE MOVES MOST OF THE HORIZON FROM "FINANCED" TO "CASH," AND THAT IS WORTH MORE THAN ANY TARIFF NEGOTIATION WE COULD WIN.***

*LEVER 2 — **HOLD THE VEHICLE LONGER.***
*DELHI, FINANCED. THE CASE ROUGHLY **TRIPLES BY YEAR 8** BECAUSE THE LOAN IS REPAID AND THE ASSET KEEPS EARNING. THIS IS THE STRONGEST SINGLE ARGUMENT IN THE DOCUMENT AND IT IS AN OPERATOR ARGUMENT — IT BELONGS TO WHOEVER OWNS THE VEHICLE.*

HORIZON	EV ₹/km	CNG ₹/km	EV ADV.	CUMMULATIVE SAVING
5yr	5.75	6.73	+14.5%	₹3,05,414
6yr	5.06	6.34	+20.1%	₹4,75,809
7yr	4.57	6.05	+24.4%	₹6,46,204
8yr	4.20	5.84	+28.0%	₹8,16,599

5. TAXES AND PURCHASE — A STRUCTURAL ADVANTAGE WORTH ₹1.5 L

	EV	CNG (Dzire Tour S)
GST	5%	18%
WHY	EV of any size or price is a separate 5% slab, no cess, retained through GST 2.0	Small car: ≤1200 cc and ≤4000 mm
Road tax — Delhi **	0% — full exemption for the life of the vehicle	~10%
Road tax — Mumbai	0% (waiver to Mar 2030)	~8%
Road tax — Bengaluru	8% (tiered from Apr 2026)	~8%

THREE CONCLUSIONS STAND OUT:

1. ***THE EV ADVANTAGE HOLDS UP AGAINST MEASURED CNG EFFICIENCY.*** *THE MEASURED BAND LEAVES AN EV ADVANTAGE OF **+19.5%** TO **+11.6%**, GIVING THE CASE A REASONABLE MARGIN UNDER THESE ASSUMPTIONS.*
2. ***ENERGY PRICES REMAIN THE REAL FRAGILITY.*** *ABOVE ROUGHLY ₹11/KWH, THE **EV LOSES ITS ADVANTAGE.** DEPOT TARIFFS OF ₹4.50–9.50/KWH SUPPORT THE CASE, WHILE PUBLIC DC CHARGING AT ₹12–24/KWH CAN OVERTURN IT.*
3. *UPTIME IS THE BINDING OPERATIONAL RISK. THE **EV HAS 45 HOURS/MONTH OF DOWNTIME** HEADROOM, AGAINST A DOCUMENTED **75 HOURS/MONTH LOST TO CHARGER QUEUES** (GREENSM). THE CASE THEREFORE DEPENDS ON RELIABLE DEPOT CHARGING, NOT JUST VEHICLE ECONOMICS*

7. WINNING CONDITIONS

CONDITION 1: CHARGING TARIFF

CHARGING ROUTE	COST	5-YR RESULT
DELHI EV TARIFF (₹4.50/KWH)	₹0.64/km	+14.5% ✓
COMMERCIAL BAND (₹8.50/KWH)	₹1.21/km	+6.1% ✓
PUBLIC DC FAST CHARGE (₹12/KWH)	₹1.71/km	−1.4% ✗

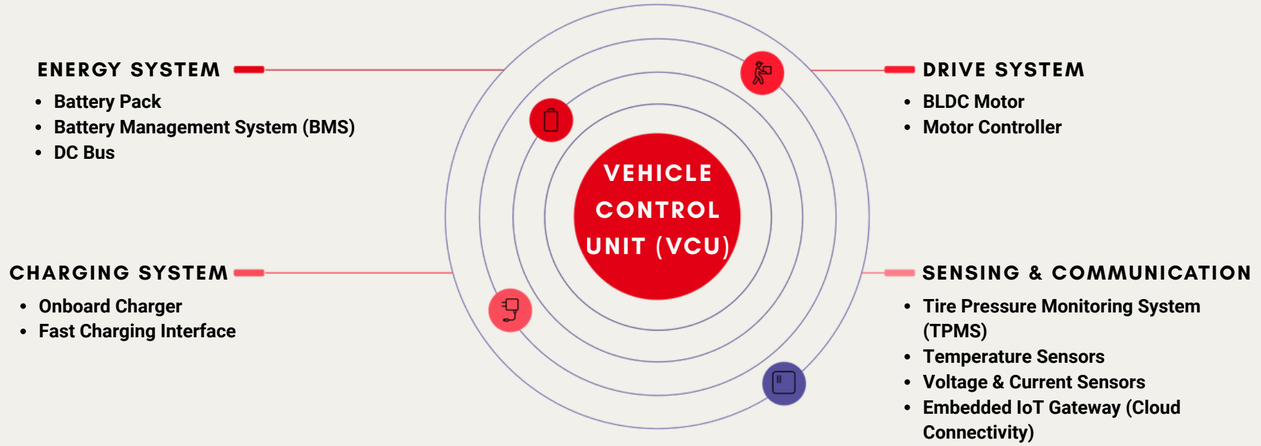
CONDITION 2: UPTIME

UPTIME METRIC	MONTHLY DOWNTIME
EV'S ALLOWABLE EXTRA DOWNTIME BEFORE CNG WINS	45 hrs
OBSERVED DOWNTIME FROM PUBLIC-CHARGER QUEUES (GREEN SM, NOIDA)	~75 hrs
HEADROOM VS OBSERVED FAILURE	0.61×

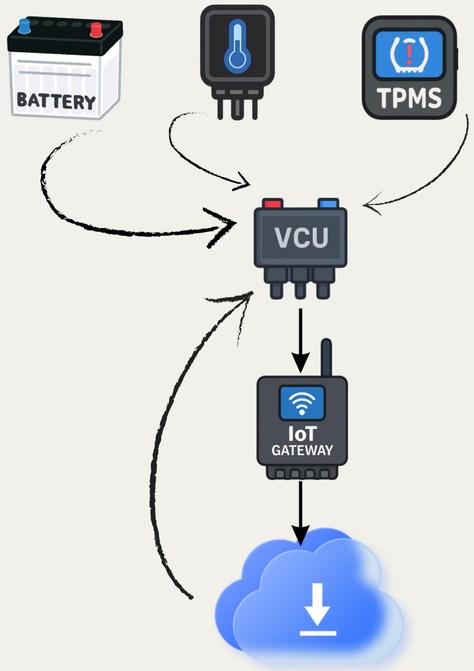
AI & CONNECTED MOBILITY ECOSYSTEM

A Control System that goes on top of an Existing EV and resolves:

1. Charging time and Battery Degradation
2. Predictive Maintenance
3. Fault Detection
4. Real Time Visibility



SYSTEM ARCHITECTURE



DATA ACQUISITION

- VCU pulls real-time data from all subsystems via CAN bus:
- BMS — SOC, SOH, cell voltage, current, temperature
- Motor controller — speed, torque, power draw
- TPMS — tyre pressure and temperature per wheel
- Charging system — charge status, current, session data
- GPS/NavIC — vehicle location

DATA PROCESSING

- Two-layer processing. Rule-based runs onboard, always. AI runs on cloud, continuously.
- Thermal protection: reduce current above 45°C, stop above 50°C
- Overvoltage / overcurrent shutdown, speed limiting below 20% SOC
- SOH estimation from charge-discharge history
- Fault detection, range prediction, adaptive charging optimisation

REAL-TIME DECISION MAKING

- VCU acts on processed data immediately. Three trigger types:
- Safety → stop charging, alert driver, log fault to cloud
- Performance → adjust torque, limit speed by condition
- Energy → optimise charging rate and timing by shift schedule
- Protection logic stays active even if cloud connectivity drops.

CONTROL EXECUTION VIA CAN BUS

Four subsystems controlled directly by VCU:

- Motor controller — torque and speed adjusted by SOC and temperature state
- Charger — current modulated by battery health; stops on unsafe condition
- BMS — continuous cell monitoring; protection activated on fault
- Dashboard — live SOC, range, system status; driver alerts on fault

CLOUD CONNECTIVITY (IOT INTEGRATION)

- MQTT for real-time telemetry, HTTP for reporting. 72-hour local buffer for network-shadow areas. Critical safety events resolved onboard, not cloud-dependent.
- Transmitted data: SOC, SOH, voltage, temperature, fault logs, TPMS readings, GPS location, charging session history.
- Storage and processing on Firebase or AWS IoT. Fleet dashboard via Grafana or equivalent.

SMART FEATURES

- Fleet Monitoring — live dashboard per vehicle: SOC, location, faults, battery status
- Predictive Maintenance — early fault detection, ≥7-day advance warning, 20–30% less downtime
- Smart Dispatch — SOC and location matched to demand, ~15% fewer deadhead kilometres
- Smart Charging — schedule built around shift demand and tariff window, 15–18% lower charging cost
- Remote Diagnostics — faults identified without physical inspection, reduces workshop time

DECISION-MAKING CORE

```
Initialize system

while(vehicle ON)
{
    // Data Acquisition
    read_BMS();
    read_motor();
    read_TPMS();
    read_sensors();

    // Safety Checks
    if(temp > 50) stop_charging();
    else if(temp > 45) reduce_current();

    if(SOC < 15) limit_speed();
    if(overcurrent) shutdown_system();

    // AI Processing
    SOH = estimate_SOH();
    range = predict_range();
    detect_faults();

    // Decision Making
    control_motor();
    control_charging();

    // Communication
    send_CAN_commands();
    send_data_to_cloud();

    delay(100ms);
}
```

MARUTI SUZUKI

PREDICTIVE MAINTENANCE

XCELerate 2026

DRIVING SUSTAINABILITY



INTELLIGENT MONITORING

- Onboard Intelligence — VCU centralises all subsystem control. BMS handles SOC estimation, cell balancing and protection. TPMS monitors tyre pressure and temperature per wheel in real time.
- AI & Predictive Layer — SOH estimation, fault prediction, range prediction and adaptive charging optimisation. Early detection targets ≥7-day advance maintenance warning.
- Safety & Thermal Protection — Overvoltage, overcurrent and over-temperature cutoffs active at all times. Thermal management protects battery life and prevents runaway.
- Connected & Scalable — IoT telemetry, fleet dashboard, remote diagnostics and CC–CV fast charging. Architecture scales from single vehicle to full depot fleet.



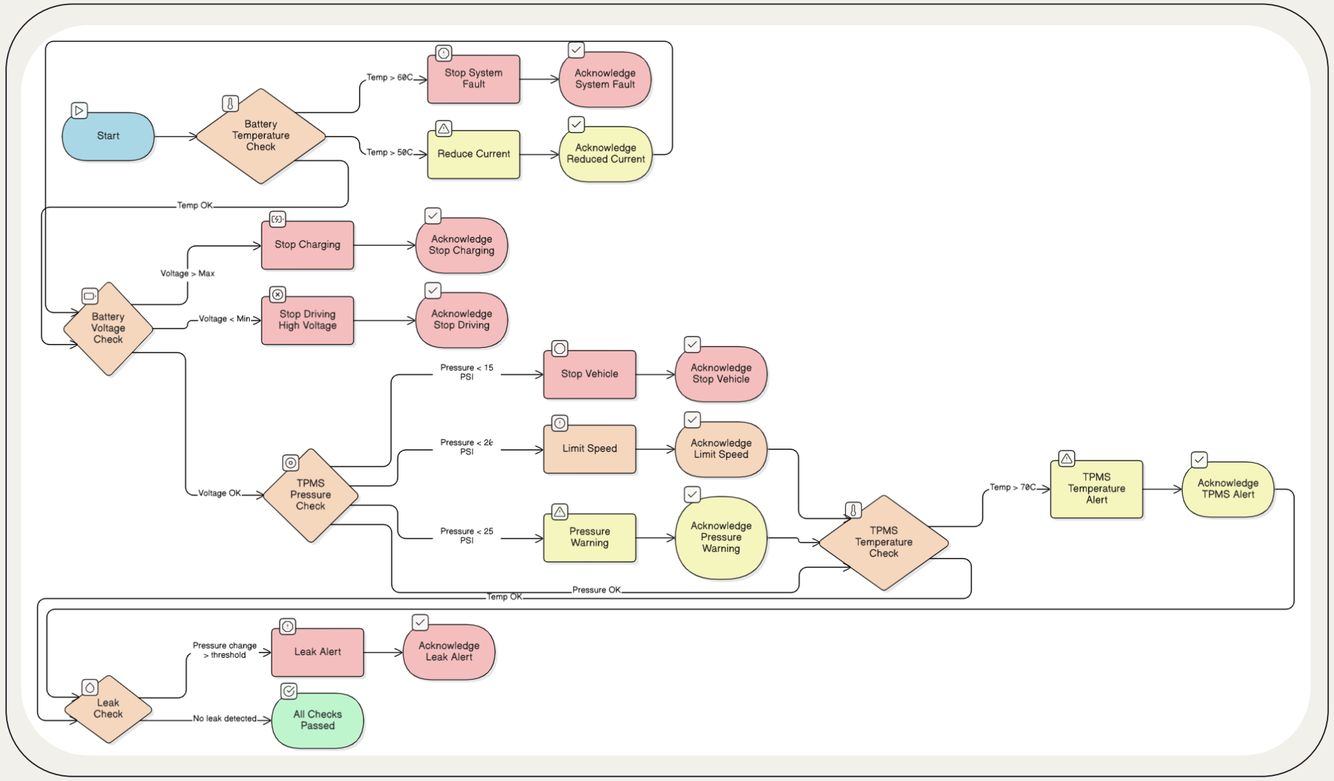
TIRE PRESSURE MONITORING SYSTEM

- Continuous Monitoring — Pressure and temperature measured per wheel in real time, displayed live on dashboard with per-wheel status.
- Three-Level Threshold — Normal: 32–35 psi, 30–50°C. Warning: below 28 psi or above 60°C. Critical: below 22 psi or above 70°C — VCU alert triggered at each level.
- Leak Detection — Rate of pressure drop monitored continuously. Rapid or gradual loss flagged before reaching critical threshold, logged to VCU and cloud.
- System Value — Correct pressure reduces rolling resistance and extends range. Early detection prevents mid-shift tyre failures on revenue-earning vehicles.

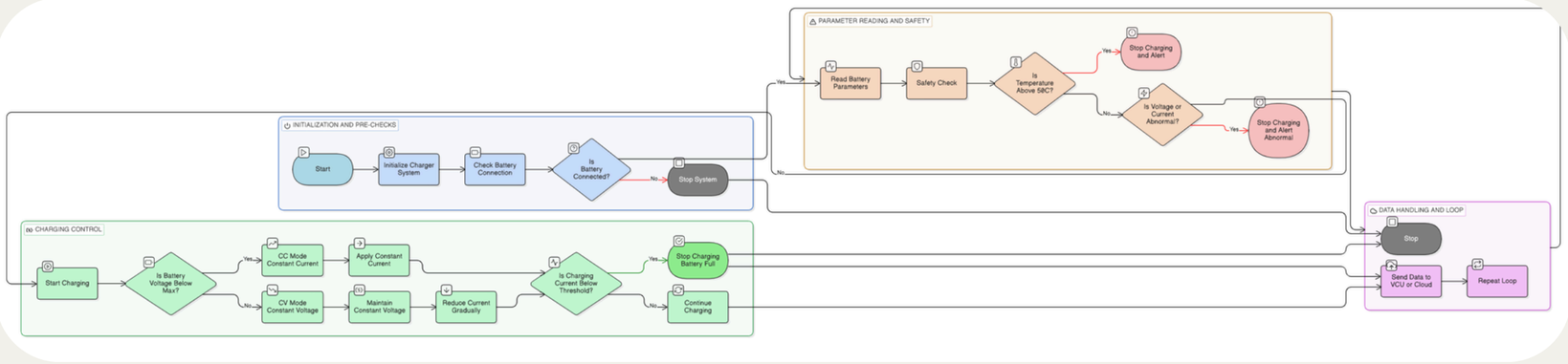


FAST CHARGING SYSTEM

- Initialization & Safety Check — System verifies battery connection, reads voltage, current and temperature. Charging stops automatically if any parameter is outside safe limits.
- CC Phase — Fixed high current delivered until battery reaches peak voltage. Charges to ~70–80% capacity quickly.
- CV Phase — Voltage held constant, current tapers down to threshold. Prevents overcharging and thermal stress on cells.
- Continuous Monitoring — Parameters tracked throughout. All safety checks remain active. Data logged to VCU and cloud in real time.



INTEGRATED BATTERY & TIRE SAFETY MANAGEMENT FLOWCHART



INTEGRATED BATTERY & TYRE SAFETY MANAGEMENT

- Battery Temperature Check — Above 45°C: charging current reduced. Above 50°C: system fault triggered, charging stopped. Prevents thermal runaway before it starts.
- Battery Voltage Check — Overvoltage stops charging immediately. Undervoltage stops vehicle operation. Both conditions acknowledged and logged before any restart.
- TPMS Pressure & Temperature Check — Below 25 psi: pressure warning issued. Below 26 psi: speed limited. Below 15 psi: vehicle stopped. Tyre temperature above 70°C triggers independent TPMS alert regardless of pressure state.
- Leak Detection & All-Clear — Rate of pressure change monitored continuously. Sudden drop beyond threshold triggers leak alert. Only when all four checks pass does the system return All Checks Passed.
- Acknowledge & Loop Architecture — Every fault state requires explicit acknowledgement before the system resets. No silent recovery. After acknowledgement, checks resume from the top of the loop — continuous protection, not one-time validation.

AI & CONNECTED MOBILITY

ROLE OF IOT IN EV SYSTEMS

IOT ENABLES REAL-TIME CONNECTIVITY AND DATA EXCHANGE BETWEEN THE VEHICLE, CLOUD, AND USER.

- COLLECT DATA FROM SENSORS (BATTERY, MOTOR, TPMS, GPS)
- TRANSMIT DATA TO CLOUD USING PROTOCOLS LIKE MQTT/HTTP
- ENABLE REMOTE MONITORING AND CONTROL
- SUPPORT FLEET MANAGEMENT SYSTEMS

ROLE OF AI IN EV SYSTEMS

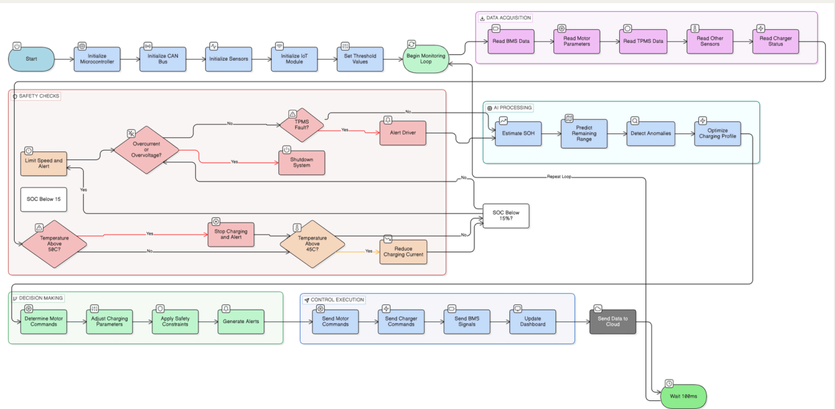
AI PROCESSES REAL-TIME AND HISTORICAL DATA TO PROVIDE INTELLIGENT INSIGHTS AND DECISION-MAKING.

- PREDICT BATTERY HEALTH (SOH ESTIMATION)
- DETECT FAULTS AND ANOMALIES
- PREDICT VEHICLE RANGE
- OPTIMIZE CHARGING AND ENERGY USAGE
- ENABLE ADAPTIVE CONTROL

INTEGRATION OF IOT AND AI

WORKING FLOW:

- SENSORS COLLECT REAL-TIME DATA
- IOT MODULE SENDS DATA TO CLOUD
- CLOUD STORES AND PROCESSES DATA
- AI ANALYZES DATA
- DECISIONS ARE MADE
- COMMANDS SENT BACK TO VEHICLE



**AN AI MONITORING LOOP THAT NEVER SLEEPS —
BECAUSE A GROUNDED VEHICLE IS LOST REVENUE.**

SENSE. THINK. ACT.

On every triggered event, the system reads four inputs – voltage, current, temperature and pressure – from the sensor layer. These feed immediately into safety checks: overvoltage, overcurrent and temperature limits are validated in sequence. If any check fails, the system bypasses AI entirely and goes straight to action – this is intentional, safety response cannot wait for cloud processing.

If safety passes, two things happen in parallel. First, fault detection runs – cell imbalance detection looks for voltage divergence across cells, sensor fault detection flags any input reading outside physical possibility, and TPMS detection checks pressure drop rate. Second, the calculations block computes SOC from current integration and SOH from the longer-term charge-discharge pattern. Both outputs are transmitted simultaneously – BLE to the driver's mobile app for real-time visibility, WiFi to cloud for fleet-level logging.

The AI layer receives this combined picture – current sensor readings, calculated SOC and SOH, fault flags – and runs two functions. Fault prediction uses historical patterns against the current state to identify anomalies that rule-based checks would miss: gradual cell degradation, subtle temperature drift, early bearing wear showing up in vibration data. Charging optimisation takes the current SOH and SOC and adjusts the charge profile – a degraded cell needs a lower CC current ceiling to avoid accelerated aging, which static CC-CV cannot account for on its own.

The action layer then resolves to one of three outputs: a warning alert for conditions approaching but not breaching thresholds, a parameter adjustment where charging current or motor torque is modified in real time, or a hard cutoff command on a critical fault. After any action, the system returns to a wait state and the entire loop restarts on the next event trigger.

